Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 30 epochs with a batch size of 128. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy by dataset.

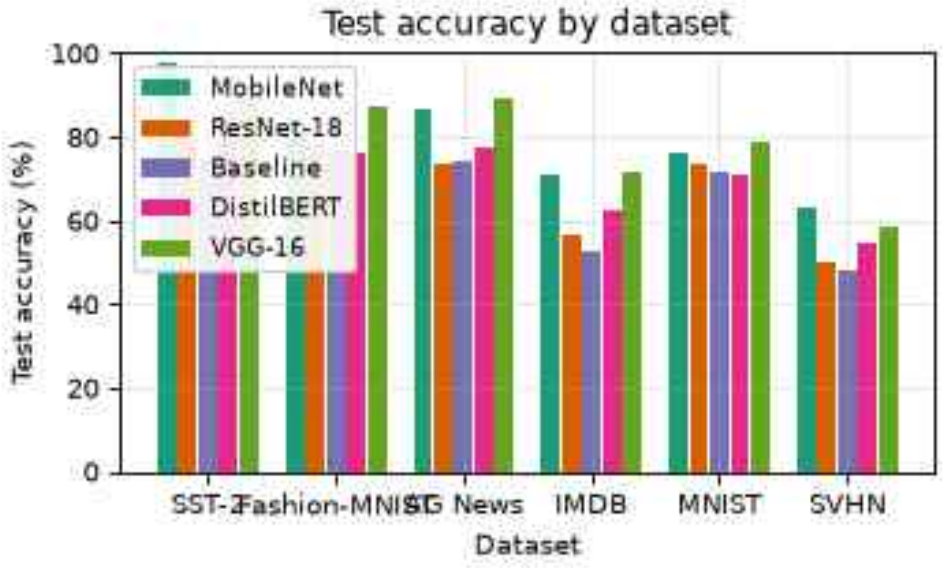


Figure 1: Test accuracy by dataset for the models in this experiment.

ResNet-18 reaches 56% test accuracy on IMDB. On AG News, VGG-16 obtains a test accuracy of about 90% (Figure 1). DistilBERT achieves 51.1% test accuracy on MNIST.

Figure 2 shows the validation accuracy during training.

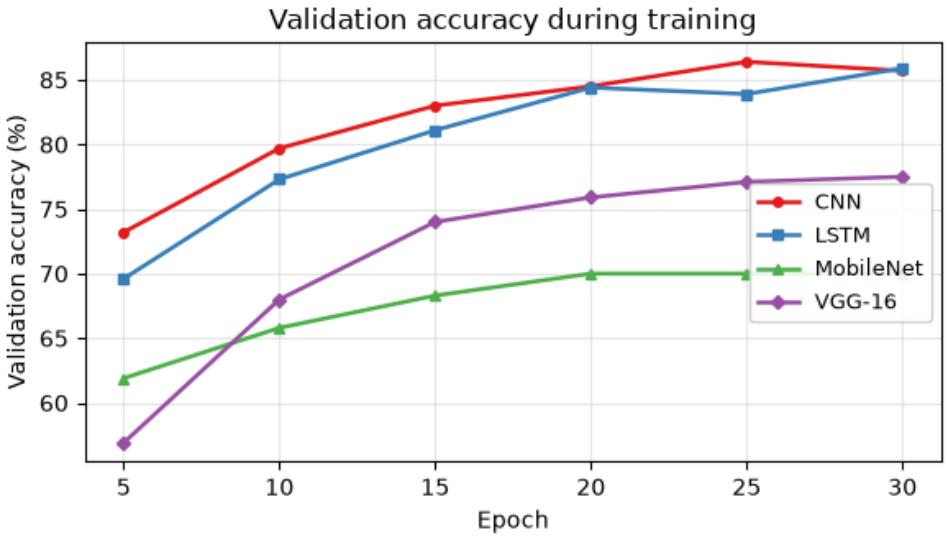


Figure 2: Validation accuracy during training for the models in this experiment.

CNN outperforms MobileNet by 16 percentage points at epoch 25 (Figure 2). From epoch 15 to epoch 25, the validation accuracy of VGG-16 improves by a relative 26.1%. LSTM attains a validation accuracy of over 80% at epoch 25.

Figure 3 shows the validation accuracy during training.

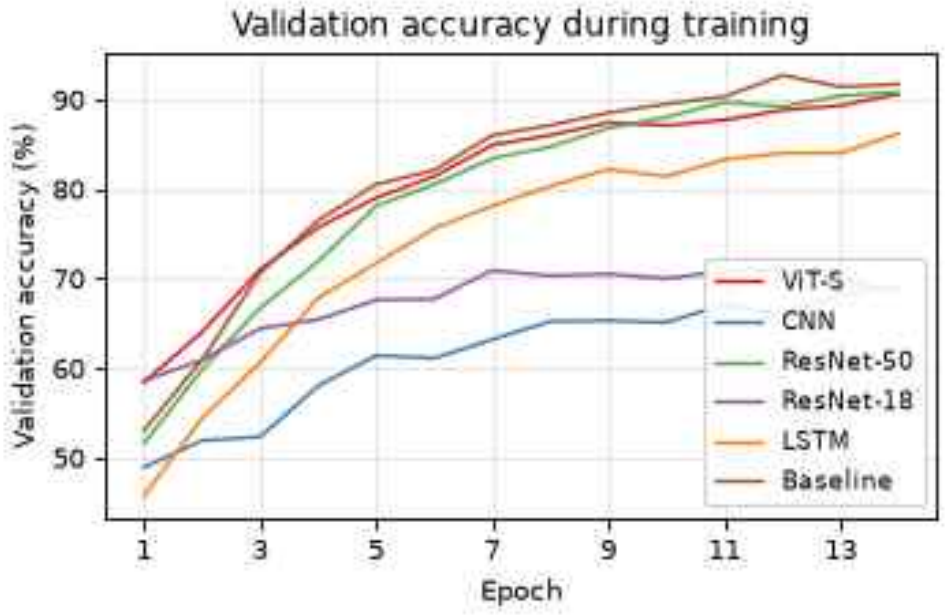


Figure 3: Validation accuracy during training for the models in this experiment.

ViT-S outperforms CNN by 21.5 percentage points at epoch 14. ResNet-18 outperforms ResNet-50 by 1.1 percentage points at epoch 2 (Figure 3). From epoch 3 to epoch 8, the validation accuracy of ViT-S improves by a relative 21.1%.

Figure 4 shows the validation accuracy of DistilBERT by dataset.

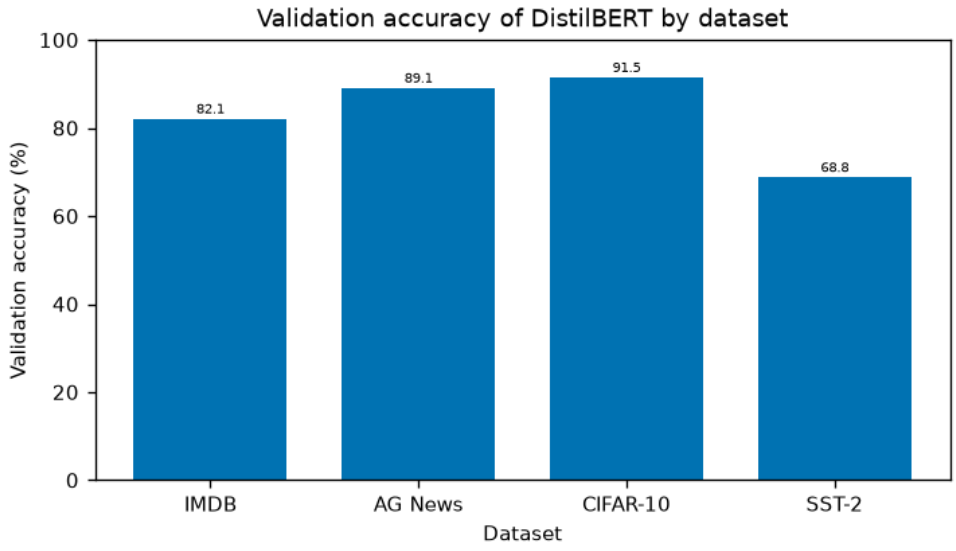


Figure 4: Validation accuracy of DistilBERT by dataset for the models in this experiment.

DistilBERT achieves 68.8% validation accuracy on SST-2. As shown in Figure 4, DistilBERT achieves 82.1% validation accuracy on IMDB. DistilBERT scores about 23 points higher on CIFAR-10 than on SST-2. On IMDB, DistilBERT obtains a validation accuracy of 82.1%.

3 Discussion

SVM reaches 52.3% accuracy on SVHN. We note that the training was stable for all models.